

Scope and Closures:

Part 1: SCOPE - What Can See What?

What is Scope?

Scope = The **visibility** and **accessibility** of variables. It answers: "From where can I access this variable?"

Think of scope like **rooms in a house**:

- Variables declared in a room are accessible in that room
 - You can look OUT from inner rooms to outer rooms
 - You CANNOT look IN from outer rooms to inner rooms
-

The Three Types of Scope

1. Global Scope (The House Itself)

Variables declared outside any function or block are **global**.

```
const globalVar = "I'm global";

function someFunction() {
  console.log(globalVar); // ✅ Can access
}

someFunction(); // "I'm global"
console.log(globalVar); // ✅ Can access from anywhere
```

Real-world analogy: Like the air outside - everyone can access it.

2. Function Scope (A Room)

Variables declared inside a function are **only accessible inside that function**.

```
function myFunction() {  
  const functionVar = "I'm in the function";  
  console.log(functionVar); // ✅ Works  
}  
  
myFunction(); // "I'm in the function"  
console.log(functionVar); // ❌ ReferenceError: functionVar is not defined
```

Key point: `var`, `let`, and `const` are all function-scoped (but `let` / `const` are also block-scoped).

3. Block Scope (A Closet in a Room)

Variables declared with `let` or `const` inside `{ }` are **block-scoped**.

```
if (true) {  
  let blockVar = "I'm in a block";  
  const alsoBlockVar = "Me too";  
  var notBlockScoped = "I'm different!";  
  
  console.log(blockVar); // ✅ Works  
}  
  
console.log(blockVar); // ❌ ReferenceError  
console.log(alsoBlockVar); // ❌ ReferenceError  
console.log(notBlockScoped); // ✅ Works! (var ignores block scope)
```

Important: `var` is NOT block-scoped, only function-scoped!

```
function testVar() {  
  if (true) {  
    var x = 10;  
  }  
  
  console.log(x); // ✅ 10 - var leaks out of block!
```

```

}

function testLet() {
  if (true) {
    let y = 10;
  }
  console.log(y); // ✗ ReferenceError - let respects block scope
}

```

Lexical Scope (The Key Concept!)

Lexical scope = Scope is determined by **where you write the code**, not where you call it.

```

const name = "Global";

function outer() {
  const name = "Outer";

  function inner() {
    const name = "Inner";
    console.log(name); // Which "name" will this print?
  }

  inner();
}

outer(); // "Inner"

```

Why "Inner"? JavaScript looks for variables in this order:

1. **Current scope** (inner function) → Found `name = "Inner"` ✓
2. If not found, check **outer scope** (outer function)
3. If not found, check **global scope**
4. If still not found → ReferenceError

This is called the **Scope Chain**.

The Scope Chain in Action

```
const level1 = "Global";

function outer() {
  const level2 = "Outer";

  function middle() {
    const level3 = "Middle";

    function inner() {
      const level4 = "Inner";

      // Can access ALL outer scopes!
      console.log(level4); // "Inner"
      console.log(level3); // "Middle"
      console.log(level2); // "Outer"
      console.log(level1); // "Global"
    }

    inner();
  }

  middle();
}

outer();
```

Visual representation of scope chain:

```
inner() scope
  ↓ (can look up)
middle() scope
```

↓ (can look up)
outer() scope
↓ (can look up)
Global scope

But you CANNOT look down:

```
function outer() {  
  function inner() {  
    const secret = "Hidden";  
  }  
  
  inner();  
  console.log(secret); // ❌ ReferenceError - can't look INTO inner function  
}
```

Part 2: CLOSURES - Functions Remember Their Birthplace

What is a Closure?

Closure = A function that **remembers** variables from its **outer scope** even after the outer function has finished executing.

This is THE most important concept for understanding JavaScript!

The Basic Closure Example

```
function outer() {  
  const message = "Hello";  
  
  function inner() {  
    console.log(message); // Accesses outer's variable  
  }  
}
```

```
    return inner; // Return the function
}

const myFunction = outer(); // outer() finishes executing
myFunction(); // "Hello" - but how does it still remember "message"?
```

What just happened?

1. `outer()` runs and creates `message`
2. `inner()` is defined INSIDE `outer()` - it "closes over" `message`
3. `outer()` returns `inner` and finishes
4. Normally, `message` would be garbage collected... **BUT**
5. `inner` still has a reference to `message` - this is a **closure**!
6. When we call `myFunction()` (which is `inner`), it still remembers `message`

Why Closures Exist

Without closures: Functions would only access their own variables and globals.

With closures: Functions can "carry" their environment with them!

```
function createCounter() {
    let count = 0; // Private variable

    return function() {
        count++; // Accesses outer variable
        return count;
    };
}

const counter = createCounter();

console.log(counter()); // 1
console.log(counter()); // 2
console.log(counter()); // 3
```

```
// "count" is NEVER directly accessible!  
console.log(count); // ❌ ReferenceError
```

Key insight: `count` lives on even though `createCounter()` finished! The returned function "closes over" it.

Real-World Example 1: Private Variables

Closures let you create **truly private** variables!

```
function createBankAccount(initialBalance) {  
  let balance = initialBalance; // PRIVATE - can't be accessed directly  
  
  return {  
    deposit: function(amount) {  
      balance += amount;  
      return balance;  
    },  
  
    withdraw: function(amount) {  
      if (amount > balance) {  
        return "Insufficient funds";  
      }  
      balance -= amount;  
      return balance;  
    },  
  
    getBalance: function() {  
      return balance;  
    }  
  };  
}  
  
const myAccount = createBankAccount(100);
```

```
console.log(myAccount.getBalance()); // 100
myAccount.deposit(50); // 150
myAccount.withdraw(30); // 120

// Can't directly access or modify balance!
console.log(myAccount.balance); // undefined
myAccount.balance = 9999999; // Doesn't work!
console.log(myAccount.getBalance()); // 120 - still protected
```

Why this works: All three methods (`deposit` , `withdraw` , `getBalance`) are closures that remember the `balance` variable!

Higher-Order Functions in JavaScript

The Simplest Definition

A higher-order function is a function that either:

1. Takes a function as an argument, OR
2. Returns a function as a result

That's it!

Why "Higher-Order"?

Think of it like hierarchy:

- **Regular values:** numbers, strings, booleans
- **First-order functions:** functions that work with regular values
- **Higher-order functions:** functions that work with other functions

In JavaScript, functions are **first-class citizens** - they can be treated like any other value (passed around, returned, stored in variables).

Type 1: Functions That Take Functions as Arguments

Example 1: Array Methods

javascript

```
const numbers = [1, 2, 3, 4, 5];

// map is a higher-order function// It takes a function as an argument
const doubled = numbers.map(function(num) {
  return num * 2;
});

console.log(doubled); // [2, 4, 6, 8, 10]
```

Why is `map` higher-order? Because it accepts a function (`function(num) { return num * 2 }`) as a parameter.

Example 2: Custom Higher-Order Function

javascript

```
// repeat is a higher-order function
function repeat(n, action) {
  for (let i = 0; i < n; i++) {
    action(i);
  }
}

// Using it:
repeat(3, function(i) {
  console.log("Iteration " + i);
});

// Output:// Iteration 0// Iteration 1// Iteration 2
```

